



NeuroAdapt: The Autonomous Neuro-Diverse Learning Ecosystem

Master Product & Research Document — v1.0

A complete end-to-end reference covering the problem, the system, the evidence, the architecture, the business model, and the roadmap.

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1. Executive Summary

NeuroAdapt is a diagnosis-agnostic, autonomous adaptive learning ecosystem designed specifically for neurodivergent students. It replaces the static, broadcast-style model of digital education with a continuously self-correcting experience that reads a learner's real-time cognitive state and reshapes every session around it.

The system operates through four tightly coupled layers: a privacy-first behavioural Observer that builds a five-dimensional State Vector every 30 seconds; a Hybrid Quantum-Classical Orchestrator that uses a Double Deep Q-Network augmented with a Variational Quantum Circuit to select one of six real-time interventions; a Generative Synthesis Engine that re-renders upcoming content into the optimal format before the learner even finishes the current slide; and a Feedback Loop that closes the reinforcement cycle through both explicit preference signals and implicit override events.

NeuroAdapt targets the approximately 20% of undergraduates who are neurodivergent — a population that is 26% less likely to complete courses and maintains significantly lower GPAs than neurotypical peers, not due to ability but due to a structural design flaw in every existing educational platform. The system is deliberately diagnosis-agnostic: it responds to behavioural signals, not clinical labels, meaning it serves the estimated 65% of neurodivergent learners who have never received a formal diagnosis and who are excluded from every conventional accommodation framework.

The global neurodivergent-focused EdTech market is valued at USD 2.4 billion in 2025 and is projected to reach USD 6.4 billion by 2032 at a CAGR of 15.1%. NeuroAdapt is positioned at the intersection of the three fastest-growing segments of that market: agentic AI, adaptive personalisation, and multimodal generative content.

2. The Problem

2.1 The Global Scale

Approximately 15 to 20 percent of the global population is neurodivergent. This umbrella term covers a range of neurological conditions where cognitive, attentional, or sensory processing diverges from the statistical norm, including:

- **ADHD** (Attention Deficit Hyperactivity Disorder) — affects approximately 5–7% of children and 2–5% of adults globally.
- **Dyslexia** — affects approximately 10–15% of the population, making it the most common learning disability.

- **Dyscalculia** — affects approximately 3–7% of the population, specifically impacting numerical processing.
- **Autism Spectrum Conditions (ASC)** — global prevalence estimated at 1 in 100 children by the WHO.
- **Dyspraxia, Tourette's, and other processing conditions** — collectively represent several additional percentage points.

In higher education specifically, studies from the US Department of Education show that 20.5% of undergraduates and 10.7% of graduate students report a disability, with ADHD and learning disabilities among the most frequently reported. This means that in any cohort of 100 students, at minimum 15 to 20 are processing the same material through fundamentally different cognitive architectures — yet every student receives an identical learning experience.

2.2 The Root Cause: A Design Flaw, Not a Student Flaw

The critical distinction that motivates NeuroAdapt is this: **the achievement gap experienced by neurodivergent students is not a product of lower cognitive potential. It is a product of a mismatched interface between the student's cognitive profile and the design of the learning environment.**

Longitudinal data establishes this clearly. College students with ADHD maintain approximately 0.5 to 0.6 grade points lower GPA across all four academic years compared to neurotypical peers. The primary driver, identified consistently across studies, is inattention — specifically the inability to sustain attention within an environment that has no mechanism to detect or respond to that inattention. The cognitive capacity is present. The platform is not designed to engage it.

This manifests in three specific failure modes:

1. Cognitive Overload. Standard digital content — dense text slides, uniform pacing, long video lectures — imposes high intrinsic cognitive load. For neurotypical learners, this is manageable. For learners with ADHD or dyslexia, the same content triggers executive function paralysis: the learner cannot process the information fast enough to make sense of it, defaults to disengagement, and falls behind. The platform interprets this as a lack of effort.

2. Executive Function Paralysis. Executive function — the set of mental skills that includes working memory, flexible thinking, and self-control — is impaired in ADHD. A learner who encounters a difficult concept does not simply "try harder." They stall. Their cursor stops moving. Their tab-switch frequency rises as their dopamine system seeks stimulation elsewhere. A traditional platform has no mechanism to detect this stall and respond with a lighter or more engaging format. It simply waits, or moves on.

3. Sensory and Format Rigidity. Dyslexic learners decode dense text at significantly higher cognitive cost than their neurotypical peers. Audio-first or visual-first learners are equally penalised by a text-dominant curriculum. The platform was designed for a hypothetical average learner that does not exist in the actual student population.

2.3 The India-Specific Compounding

In India, the failure of the education system to serve neurodivergent learners is compounded by three structural factors:

Economic exclusion. Specialised schooling, intervention centres, and occupational therapy for neurodivergent learners costs between Rs. 2.5 lakh and Rs. 8 lakh per year. This cost falls almost entirely on the family. The majority of Indian families cannot sustain this, meaning that students who do not benefit from mainstream instruction simply fall out of the system with no alternative pathway.

Diagnosis as a prerequisite. Every formal accommodation — from extended exam time to access to special educators — requires a clinical diagnosis. A formal neuropsychological assessment typically costs Rs. 15,000 or more and requires access to a qualified specialist, which in most of India means a major metro city. For the millions of students in tier-2 cities, rural areas, or lower-income households, this prerequisite is effectively an exclusion from support by another name.

Social stigmatisation. The act of seeking a diagnosis in India still carries significant social stigma in many communities. Parents and students frequently avoid the process entirely, meaning the student enters higher education with unacknowledged neurodivergence, no institutional support, and no vocabulary to describe what they are experiencing. Burnout, social isolation, and dropout follow.

NeuroAdapt eliminates all three barriers simultaneously. It requires no diagnosis, no clinical label, no prior assessment, and no institutional accommodation pathway. It observes behaviour and adapts. The support is built into the learning infrastructure itself.

2.4 What Existing Solutions Get Wrong

Current adaptive EdTech platforms fall into two categories, both of which are insufficient for neurodivergent learners.

Category A: Accessibility Features. These include closed captions, screen readers, dyslexia-friendly fonts, and high-contrast modes. These are static, opt-in features that require the student to already understand their own needs and actively configure the platform. They do not respond to the student's real-time cognitive state, they do not adapt pacing or complexity, and they do nothing for executive function regulation or cognitive load management.

Category B: Adaptive Difficulty Systems. Most modern EdTech platforms include some form of adaptive difficulty — a system that makes questions harder or easier based on whether the student got the last one right. This is a meaningful improvement over static instruction, but it optimises for a single metric (correctness) and does not account for the qualitative experience of the learner. A student who is in executive function paralysis may answer several questions incorrectly not because the content is too hard, but because they cannot sustain enough attention to process the question. An adaptive difficulty system will simply make the questions easier. It will not detect the paralysis, insert a sensory break, or re-render the content into a short video.

Neither category addresses the fundamental need: **a system that models the learner's cognitive state in real time and uses that model to make holistic decisions about content format, pacing, complexity, and sensory load simultaneously.**

3. Vision & Philosophy

NeuroAdapt is built on four foundational philosophical commitments.

1. The platform adapts to the learner; the learner does not adapt to the platform.

This inverts the implicit assumption of every educational technology built to date.

The standard model asks the student to perform within a fixed structure and blames the student when they cannot. NeuroAdapt takes the opposite position: the student's cognitive profile is fixed and sovereign; the platform's job is to serve it.

2. Neurological diversity is an asset, not a liability.

ADHD learners who reach hyperfocus states demonstrate extraordinary depth of engagement and creative problem-solving. Dyslexic learners frequently exhibit stronger spatial reasoning and holistic thinking than their peers. Autistic learners often develop exceptional domain expertise and pattern recognition. NeuroAdapt is not a remediation tool. It is an amplification tool — designed to create conditions in which each neurological profile can express its strengths.

3. Support should be invisible and universal.

Accommodation systems that require the student to self-identify and formally request support are participation barriers, not support systems. NeuroAdapt's support is invisible — woven into the fabric of the learning experience itself — and universal, meaning it applies to every learner regardless of whether they have ever been diagnosed with anything.

4. Privacy is not a compliance checkbox; it is a design principle.

Neurodivergent learners are a historically stigmatised group. Any system that surveils their learning behaviour must be held to the highest possible privacy standard. NeuroAdapt computes all sensitive signals locally in the browser. No raw interaction data ever leaves the device. This is not a policy decision made after the system is built. It is a constraint that shapes every architectural choice.

4. Target Users

NeuroAdapt is designed for the following primary user groups:

Primary: Neurodivergent Students (diagnosed and undiagnosed)

Students across higher education who experience ADHD, dyslexia, dyscalculia, autism spectrum conditions, anxiety-related attention difficulties, or any other condition

that makes standard digital learning significantly more effortful than it needs to be. Critically, NeuroAdapt serves both those who have a formal diagnosis and those who have never been assessed. The system responds to behaviour, not labels.

Secondary: Neurotypical Students Under Cognitive Load

Any student who is fatigued, anxious, under time pressure, or encountering particularly dense material will exhibit signals that NeuroAdapt recognises and responds to. The system does not require a neurodivergent profile to be useful — it simply becomes especially valuable for those whose cognitive experience diverges most consistently from the default format.

Tertiary: Educators and Institutions

Educators gain access to aggregated (not individualised) insights into how their content is being received: which sections trigger the highest stall rates, which formats drive the best completion, and which concept sequences produce the most disengagement. These insights are actionable for curriculum design without requiring any individual student's data to be exposed.

Deployment Context

NeuroAdapt is designed as a B2B platform deployed by educational institutions. The primary integration pathway is via LTI 1.3, meaning it installs as a plugin inside an existing Moodle, Canvas, or similar LMS without requiring the institution to replace its infrastructure.

5. Literature Review & Evidence Base

5.1 Neurodivergent Academic Outcomes

The evidence that neurodivergent students face systematically worse academic outcomes under current educational structures is robust and multi-layered.

Langberg et al. (2021) followed 201 college students with ADHD and 205 matched controls across four years. ADHD students maintained significantly lower GPAs from year one onward and made slower academic progress regardless of accommodation provision. The primary predictor of underperformance was inattention severity —

not motivation, not intelligence, not socioeconomic status.

Breaux et al. (2021) established that of unmedicated ADHD students, only 49% completed eight semesters versus 59% of non-ADHD peers — a gap that persisted even after controlling for demographic variables. Medicated students showed partial improvement (54%), but still lagged significantly.

Doyle et al. (2024) conducted a rapid review of support structures for neurodivergent students in higher education and found that existing accommodation frameworks — extended deadlines, note-taking support, quiet exam rooms — showed limited measurable impact on GPA and completion. The study concluded that surface-level accommodations are insufficient and that systemic platform-level redesign is required.

At the population level, data from the US National Center for Education Statistics shows that 20.5% of undergraduates report a disability, with ADHD and learning disabilities among the most prevalent categories. Adjusting for under-reporting and undiagnosed conditions, the true prevalence of neurodivergence in higher education is estimated at 20% or higher.

5.2 Adaptive Learning Technologies

Maran et al. (2024) conducted a study on optimising cognitive load and learning adaptability using adaptive technologies. The study confirmed that systems which adjust difficulty and content complexity in response to real-time performance signals reduce cognitive overload and improve both retention and on-task behaviour, particularly for learners starting from weaker baselines.

A systematic review of AI-driven intelligent tutoring systems (Al-Turki et al., 2025) found that adaptive platforms consistently outperform static instruction in test accuracy, engagement, and learner confidence, with the largest effect sizes observed in students who struggled most under traditional instruction — exactly the population NeuroAdapt targets.

5.3 Reinforcement Learning in Education

Clément et al. (2015) demonstrated in a seminal study that Multi-Armed Bandit algorithms applied to the sequencing of learning activities outperform

expert-designed static sequences in real classrooms. In a live deployment with over 400 primary school children, bandit-driven sequencing produced higher mastery rates and significantly faster learning compared to the curriculum designed by experienced teachers. Students were kept in their Zone of Proximal Development automatically, without any teacher needing to monitor individual progress.

This finding has been replicated and extended in numerous subsequent studies: RL-based ITS systems have produced 30% reductions in hint requests, higher post-test scores, and measurably higher engagement compared to non-adaptive rule-based tutoring systems. The core mechanism — sequential decision-making over content and difficulty, optimising a learning-signal reward — is directly applicable to NeuroAdapt's Orchestrator layer.

The key extension NeuroAdapt makes is to the reward function itself: existing RL tutoring systems optimise academic performance metrics (correctness, completion). NeuroAdapt's Stability Reward incorporates cognitive and affective signals — penalising overload states and rewarding stable engagement — which is specifically appropriate for neurodivergent learners whose performance metrics often reflect cognitive state rather than knowledge.

5.4 Gamification & Multimodal Interventions for ADHD

The strongest clinical evidence for technology-based interventions for ADHD comes from digital therapeutics research, particularly the EndeavorRx program (AKL-T01). EndeavorRx is an FDA-authorised prescription video game treatment for ADHD in children aged 8–12. Across five clinical studies involving over 600 participants, EndeavorRx demonstrated:

- Statistically significant improvements in sustained and selective attention as measured by the Test of Variables of Attention (TOVA).
- Approximately one-third of treated children moving into the normative range on at least one objective attention measure.
- A mean TOVA Attention Performance Index improvement of 0.93 versus control in the pivotal RCT.
- 73% of children reporting subjectively improved attention after four weeks of daily play at 25 minutes per session.

Leitner et al. (2024) replicated these findings in a PMC-published RCT, confirming that game-based digital interventions produce clinically meaningful improvements in attention and executive function in ADHD populations.

Teruel et al. (2024) demonstrated that biometric signals captured passively during a computer game — without any explicit testing — can accurately measure attentional performance in ADHD patients, validating the Observer model that NeuroAdapt uses to infer cognitive state from passive behavioural data.

Frontiers in Education (2025) published findings confirming that gamified educational applications produce significant improvements in both sustained attention and academic performance in ADHD children compared to conventional instruction.

5.5 Hybrid Quantum-Classical Models

Bhatia et al. (2025) demonstrated that Variational Quantum Circuits outperform equivalent classical dense layers in machine learning tasks, with the primary advantage being faster convergence — specifically, significantly lower loss values in the early training epochs compared to classical baselines of similar parameter count.

Cao et al. (2025) confirmed this advantage in hybrid convolutional architectures published in Nature Scientific Reports, establishing that the primary benefit of VQCs is not marginal accuracy improvement at convergence, but measurably faster learning during the initial training phase.

Both findings directly support NeuroAdapt's use case. The system does not need a model that eventually becomes marginally more accurate than a classical baseline after thousands of sessions. It needs a model that builds an accurate per-learner profile in the first three to five sessions — before the student has had the chance to disengage. The VQC's early-convergence advantage is precisely the property needed.

6. Objectives

NeuroAdapt is built around three core, measurable goals:

Objective 1 — Close the Achievement Gap

Enable real-time, personalised learning environments targeting the approximately 20% of undergraduates who are neurodivergent, with measurable outcomes of a 25% improvement in module completion rates and a 10–15% gain in assessment performance. Contribution to reducing the documented dropout gap between neurodivergent and neurotypical students in higher education.

Objective 2 — Shift from Accessibility to Stability

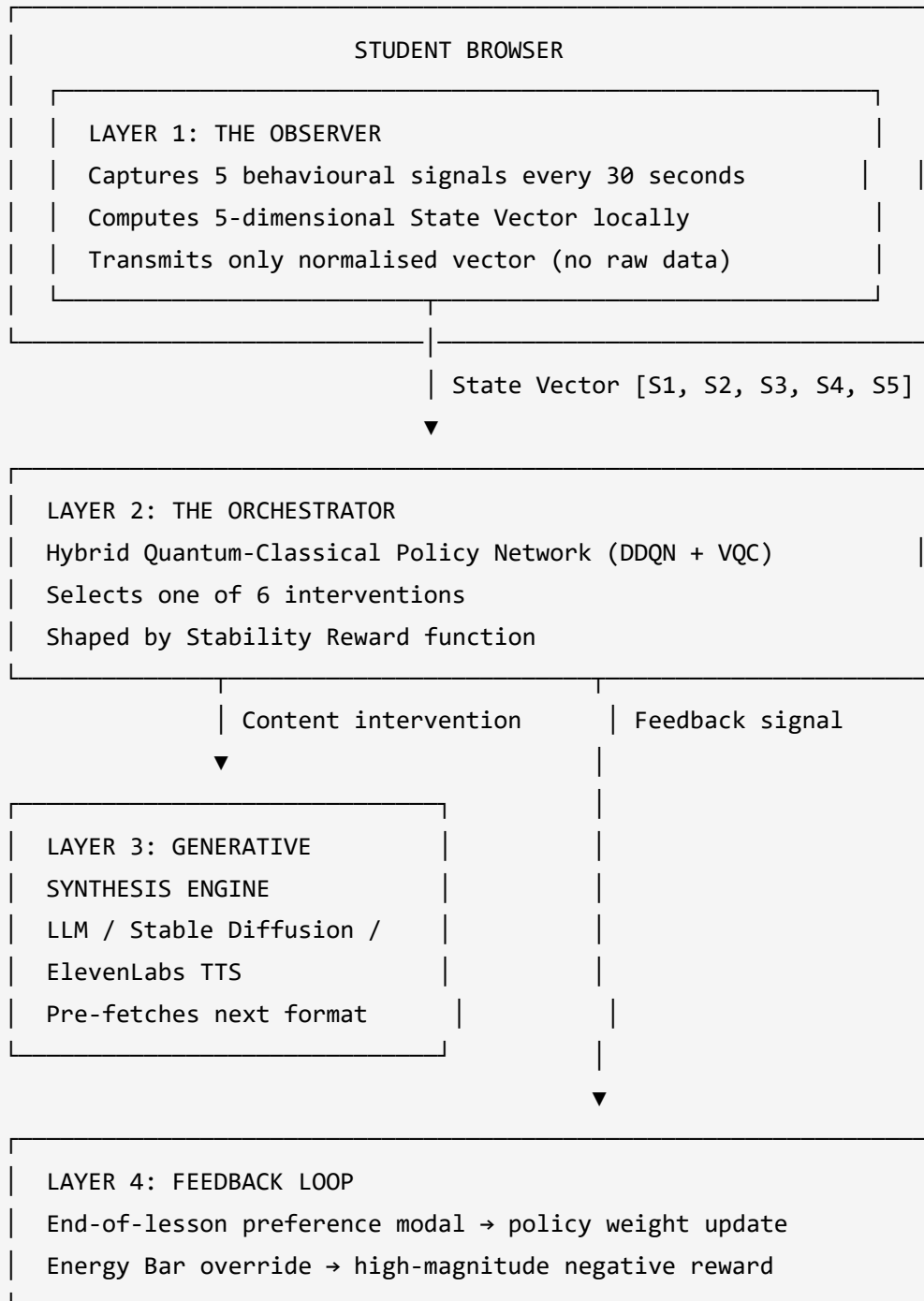
Move beyond surface-level accommodations toward a Stability-Focused model. The system does not provide closed captions or font settings and call it accessibility. It actively manages the student's cognitive load in real time, continuously measuring whether the learner is in an optimal engagement state and intervening immediately when they are not.

Objective 3 — Serve the Undiagnosed Majority

Because NeuroAdapt responds to behavioural signals rather than clinical labels, it provides adaptive support to the estimated 65% of neurodivergent students who have no formal diagnosis and who are currently excluded from every institutional support framework. This is both the most ambitious and the most practically impactful of the three objectives.

7. System Architecture Overview

NeuroAdapt's architecture consists of four tightly coupled layers that together transform a passive digital learning session into a continuously self-correcting experience.



This architecture has three fundamental properties:

1. **It is closed-loop.** The system continuously learns from its own interventions. Every decision the Orchestrator makes is evaluated against the learner's subsequent state, and the policy updates accordingly.
2. **It is privacy-preserving by construction.** The most sensitive data — raw

interaction patterns — never leave the browser. Only a normalised, anonymised state vector is transmitted.

3. **It is modality-agnostic.** The Orchestrator selects what kind of experience to deliver. The Synthesis Engine executes the delivery. These two concerns are deliberately separated, meaning the content generation stack can be upgraded (swap ElevenLabs for a different TTS provider, swap Stable Diffusion for a newer image model) without touching the policy layer.
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8. Layer 1 — The Observer

8.1 Why Behavioural Signals, Not Biometrics

The most accurate real-time measure of cognitive state would be physiological: pupil dilation, galvanic skin response, heart rate variability, EEG. Research confirms that these signals can predict cognitive load and emotional state with high accuracy. NeuroAdapt does not use any of them.

The reasons are practical and ethical. Physiological sensing requires hardware that most students do not own and would not consent to wearing during routine study sessions. Camera-based sensing (eye tracking, facial expression analysis) requires always-on video, which is both technically costly and a profound privacy violation for a population that already faces stigma. Even if consent were obtained, the regulatory and ethical risk of storing such data about a neurodiverse population is significant.

Instead, NeuroAdapt uses five passive behavioural signals that can be captured entirely within the browser, on the student's existing device, with zero additional hardware, and without any camera or audio input. These signals are computed locally and never transmitted in raw form.

8.2 The Five Signals

Signal 1 — Semantic Dwell Ratio (SDR)

Definition: Time spent on a content element divided by the word count or visual

complexity of that element.

What it measures: Whether the learner is spending time proportional to the content's actual density. A very low SDR on a dense paragraph indicates impulsive skimming — the learner is scrolling past content they have not processed. A very high SDR indicates a "reading wall" — the learner is stuck, re-reading, or unable to progress. Both are signals of potential mismatch between content format and cognitive state.

Why it matters for neurodivergence: Dyslexic learners typically show elevated SDR on text-heavy content because decoding takes longer. ADHD learners show characteristically low SDR followed by sudden stalls — a pattern no simple "time on page" metric can capture.

Signal 2 — Interaction Jitter (IJ)

Definition: The variance of mouse velocity (or touch input velocity) over a rolling 30-second window.

What it measures: The smoothness or choppiness of the learner's interaction with the interface. High, erratic variance indicates agitation, anxiety, or restlessness. Very low variance over an extended period indicates either very calm focus or complete disengagement (stall).

Why it matters for neurodivergence: Anxiety frequently co-occurs with ADHD, dyslexia, and autism. High jitter is a reliable proxy for an elevated anxiety or restlessness state. When combined with other signals, it helps distinguish engaged hyperfocus (high focus + moderate jitter) from anxious struggling (high jitter + low dwell + high stall duration).

Signal 3 — Focus Persistence (FP)

Definition: The frequency of tab switches or application context changes within a session window.

What it measures: The learner's ability to maintain attention on the current learning context. Every tab switch represents a break in attentional continuity — the learner's dopamine system has sought stimulation elsewhere.

Why it matters for neurodivergence: Tab-switching frequency is one of the most reliable behavioural proxies for ADHD inattention in a digital environment. Neurotypical learners under normal conditions switch context far less frequently than those experiencing attentional dysregulation. A sharp increase in tab switching during a session indicates the content format has stopped holding attention and an intervention is needed.

Signal 4 — Stall Duration (SD)

Definition: The continuous time elapsed since the learner's last meaningful interaction with the content (scroll, click, keypress, or media interaction).

What it measures: Whether the learner has stopped engaging. Critically, stall duration alone is ambiguous — it could indicate deep thinking (the learner is processing a complex idea and doesn't need to scroll) or executive function paralysis (the learner is overwhelmed and cannot initiate the next action).

Why it matters for neurodivergence: Executive function paralysis — the inability to initiate a task even when one intends to — is one of the most debilitating symptoms of ADHD and is poorly understood by neurotypical observers who interpret it as laziness. NeuroAdapt disambiguates stall duration from other signals: a stall accompanied by high jitter suggests anxiety-driven paralysis. A stall accompanied by low jitter and normal SDR suggests active contemplation. The appropriate intervention differs dramatically.

Signal 5 — Preference Delta (PD)

Definition: The explicit format choice made by the learner at the end of each lesson module, captured via a single-question modal.

What it measures: Ground truth. The learner knows, in retrospect, which format they found most useful during that session. This signal serves as the reward label that closes the reinforcement learning cycle.

Why it matters for neurodivergence: While signals 1–4 are inferred proxies for cognitive state, Signal 5 is direct self-report. It grounds the entire policy in the learner's own experience rather than purely in behavioural inference. The gap between what the Orchestrator predicted would be most useful and what the learner reports was most useful is the primary convergence metric for the system.

8.3 The State Vector

At every 30-second polling interval, the five signals are normalised to a continuous [0, 1] range and assembled into a five-dimensional State Vector:

```
S = [SDR_norm, IJ_norm, FP_norm, SD_norm, PD_prev]
```

Where `PD_prev` is the Preference Delta from the most recent completed session (carried forward as a prior). This vector is the only data transmitted to the server and is the sole input to the Orchestrator.

8.4 Privacy & Compliance

All five signals are computed on-device using browser JavaScript. The raw events (individual mouse coordinates, specific scroll timestamps, tab-switch times) are processed locally and immediately discarded. Only the normalised aggregate values are retained and transmitted. This means:

- No raw interaction data or personally identifiable information ever leaves the client device.
- The server never sees which specific URLs were visited during a tab switch.
- The server never sees the specific scroll patterns or click coordinates that produced the normalised value.

This architecture satisfies the data minimisation requirements of India's Digital Personal Data Protection Act (DPDP Act, 2023) by design. It also aligns with GDPR data minimisation principles for deployments in European jurisdictions.

9. Layer 2 — The Orchestrator

The Orchestrator is the decision-making core of NeuroAdapt. It receives the five-dimensional State Vector from the Observer and outputs a single integer from 0 to 5 representing the selected intervention. This decision is made by a Hybrid Quantum-Classical Policy Network trained via Double Deep Q-Learning (DDQN) with a Stability Reward function.

9.1 The Six Interventions

The action space of the Orchestrator consists of six possible interventions, ordered from least to most disruptive:

Action 0 — Hold Course

No change to the current content format or pacing. The learner is in an optimal or near-optimal cognitive state. The platform continues the current lesson without interruption. This is the most common action during well-matched sessions.

Action 1 — Soft Nudge

A brief, unobtrusive re-engagement prompt — a subtle animation, a progress indicator, or a micro-question — that brings the learner's attention back to the content without disrupting the flow or signalling that anything is wrong. Used when signals indicate mild inattention that has not yet escalated.

Action 2 — Simplify Text

The Generative Synthesis Engine is triggered to reduce the lexical complexity and sentence density of the next content segment. An LLM rewrites the upcoming material at a lower reading level while preserving conceptual accuracy. Used when SDR is very high (learner is stuck on dense text) or when stall duration is moderate and jitter is low.

Action 3 — Switch to Video

The Generative Synthesis Engine produces or retrieves a short video rendition of the upcoming content — either a pre-synthesised avatar explanation (HeyGen) or an AI-generated visual narrative (Stable Diffusion + voiceover). Used when text-based

formats have consistently produced poor engagement for this learner, or when real-time signals indicate the learner needs a different sensory modality.

Action 4 — Inject Gamified Task

A short, adaptive challenge burst is injected into the session — a quick-fire question set, a visual matching task, or a decision-tree scenario — designed to re-engage the learner's dopamine system through novelty and agency. Used when Focus Persistence is very low (high tab-switching) and Stall Duration is escalating. Based on clinical evidence (EndeavorRx, Leitner 2024) that gamified adaptive tasks produce significant attention recovery in ADHD profiles.

Action 5 — Sensory Break

The session is paused. A guided breathing or grounding exercise is offered, or the learner is simply invited to step away for a defined period. This is the most significant intervention and is used only when multiple signals simultaneously indicate that the learner is in executive function paralysis or sensory overload: very high stall duration, very high jitter, very low dwell ratio, and very low focus persistence.

9.2 The Stability Reward Function

The reward function is the most philosophically significant design decision in the entire NeuroAdapt system. Conventional RL tutoring systems reward correctness: the policy gets positive reward when the student answers a question correctly. This is appropriate for neurotypical learners but is actively harmful for neurodivergent learners, because it optimises for performance without regard for the cognitive experience that produces it.

NeuroAdapt's Stability Reward is shaped by three components:

Engagement Stability Term (positive): Reward is proportional to the degree to which the learner's state vector remains within a defined "optimal zone" — low stall duration, moderate dwell ratio, low jitter, stable focus persistence. A learner who maintains stable engagement across a session earns maximum reward for the Orchestrator.

Overload Penalty (negative): The reward is negatively penalised when the

state vector indicates cognitive overload: simultaneously high stall duration and high jitter. This teaches the Orchestrator that allowing a learner to persist in an overloaded state is worse than intervening, even if the intervention interrupts the flow of the lesson.

Energy Bar Override Term (high-magnitude negative): When the learner manually triggers the Energy Bar (described in Layer 4), this fires an extremely large negative reward signal into the replay buffer. This teaches the Orchestrator that it missed a build-up of stress that the learner themselves had to arrest. Across multiple such events, the Orchestrator learns to intervene preventively, earlier in the signal progression, so that the learner never reaches the point of needing to manually override.

The Preference Delta from the end-of-lesson modal is used as the explicit reward label for the format selection component: when the learner reports that the format chosen by the Orchestrator matched their preference, the policy receives positive reinforcement for that state-action pair.

9.3 The Hybrid Quantum-Classical Policy Network

The policy network that drives the Orchestrator is a Double Deep Q-Network (DDQN) with a critical architectural modification: the inner hidden layer is replaced by a Variational Quantum Circuit (VQC) implemented in PennyLane.

The overall architecture is:

```
Input Layer (5 neurons, one per state dimension)
    ↓
Classical Dense Layer (encoding and normalisation)
    ↓
Variational Quantum Circuit (inner hidden layer)
    ↓
Classical Dense Layer (output decoding)
    ↓
Output Layer (6 neurons, one per action)
```

The VQC layer:

1. **Encodes** each of the five state features as a rotation angle applied to a dedicated qubit using Rx, Ry, or Rz gates.
2. **Entangles** the five qubits via a structured layer of CNOT gates designed to capture correlations between signal pairs that are diagnostically meaningful.
3. **Measures** each qubit in the Z basis, producing five expectation values in the range $[-1, 1]$ that are passed to the classical output decoder.

The trainable parameters of the VQC are the rotation angles in the variational layers, optimised jointly with the weights of the classical layers via standard gradient descent through the parameter-shift rule.

9.4 Why a Variational Quantum Circuit

The decision to use a VQC is justified on two grounds: expressivity and convergence speed.

Expressivity. Five classical binary inputs produce $2^5 = 32$ distinct combinatorial states. A classical dense layer maps these 32 states into a higher-dimensional feature space through learned weights. A VQC maps the same five inputs into a $2^5 = 32$ -dimensional Hilbert space through quantum superposition, exploring this space simultaneously rather than sequentially. With five entangled qubits, the VQC can represent correlations between all possible subsets of the five signals in a single forward pass — a property called quantum parallelism.

Convergence speed. The most important property for NeuroAdapt's use case is not final accuracy but early convergence. The system must build a useful model of each individual learner within the first three to five sessions — before the learner disengages. Literature on hybrid quantum-classical architectures (Bhatia et al. 2025; Cao et al. 2025) confirms that VQCs consistently reduce loss more rapidly in early training epochs compared to classical baselines of equivalent parameter count. The system essentially "learns the learner" faster.

Parameter efficiency. The VQC achieves its representational power with significantly fewer parameters than an equivalent classical hidden layer, which keeps inference overhead low enough for real-time decision-making.

9.5 ADHD Hyperfocus Detection via Entanglement

A specific design decision in the CNOT gate structure of the VQC is the explicit entanglement of the Focus Persistence qubit with the Interaction Jitter qubit.

This pair captures a cognitively important but counterintuitive state: the ADHD hyperfocus state, characterised by simultaneous **high focus** (low tab-switching, high dwell) and **high jitter** (restless, energetic mouse movement). This is the opposite of the typical overload signal (high jitter + low focus). A classical dense layer processes these two signals independently and would likely misclassify hyperfocus as mild agitation. The CNOT entanglement between these two qubits means the VQC natively represents the correlation between them, allowing it to correctly identify hyperfocus as a positive engagement state that the Orchestrator should not interrupt.

10. Layer 3 — The Generative Synthesis Engine

The Generative Synthesis Engine executes the content interventions selected by the Orchestrator. It is responsible for producing or retrieving the appropriate content format and delivering it to the learner before — or at the moment — the current content segment ends.

10.1 Text Re-rendering

When the Orchestrator selects Action 2 (Simplify Text), the Synthesis Engine calls an LLM (GPT-4o in the production configuration, Llama-3 for open-source deployment) with the original content segment and a prompt specifying the target reading level and structural simplifications required:

- Reduction of average sentence length
- Replacement of technical jargon with plain-language equivalents (with definitions preserved in tooltips)
- Breaking of paragraph-dense sections into bullet-pointed summaries
- Addition of bolded key terms to aid scanning

The re-rendered text preserves the conceptual accuracy and curriculum alignment of the original. It does not simplify the underlying concepts — only the linguistic presentation.

10.2 Visual Synthesis

When the Orchestrator selects Action 3 (Switch to Video), the Synthesis Engine produces a short video segment in one of two modes:

Avatar Video (HeyGen): A human-presented avatar narrates the key concept in a calm, structured, direct format. The script is generated by the LLM from the original content. Tone and pacing are calibrated for clarity rather than entertainment. For autism and high-anxiety profiles, the avatar speaks slowly, makes no sudden movements, and uses no distracting background elements.

Illustrated Narrative (Stable Diffusion + voiceover): For concepts that benefit from visual analogy rather than direct explanation, the Synthesis Engine generates a sequence of illustrative images with AI-generated voiceover narration. This is particularly effective for abstract mathematical or scientific concepts that learners with dyscalculia or spatial processing differences find easier to grasp through visual metaphor.

10.3 Audio Synthesis

When audio is delivered without video — as a secondary channel alongside simplified text, or as part of a sensory break — the Synthesis Engine uses ElevenLabs TTS (or Coqui TTS in the open-source configuration) with a specific voice preset:

- Calm, neutral, gender-neutral voice tone
- Slow to moderate speaking pace
- Minimal prosodic variation (avoids sudden emphasis or pitch changes that can be overstimulating for autism spectrum or high-anxiety learners)
- No background music or ambient sound

The audio preset is configurable by the educator at the institutional level and can be overridden by the learner at the individual level.

10.4 Predictive Pre-fetching

A key performance feature of the Synthesis Engine is its predictive pre-fetching mechanism. Rather than generating the next format variant only after the Orchestrator makes its decision, the engine begins generating all likely next-format variants asynchronously while the learner is still on the current slide.

The pre-fetch priority is determined by the Orchestrator's Q-value distribution: the top two or three actions by Q-value each trigger a pre-fetch of their corresponding content variant. When the Orchestrator makes its final decision, the winning format variant is already generated and ready to deliver. From the learner's perspective, the transition to a new format appears instantaneous.

10.5 The Cost Gate

Generative API calls — particularly to GPT-4o and HeyGen — represent the only significant variable cost in the NeuroAdapt system. Without constraint, a system that generates new content every 30 seconds would be economically unviable at scale.

NeuroAdapt addresses this through a confidence threshold gate: generative API calls are only executed when the Orchestrator's selected action is not Action 0 (Hold Course) and the Q-value for that action exceeds a threshold of 0.60 (i.e., the system is at least 60% confident that a format change is needed). When the learner is in a stable engagement state, the system holds course, no API calls are made, and the cost for that session is effectively zero.

Empirically, well-matched learners who are in flow spend the majority of their session in Action 0. API calls are made only for struggling sessions, where the cost is more than justified by the outcome improvement.

For institutions with hard cost constraints, every paid API component has a zero-cost open-source substitute: Llama-3 (self-hosted or via Groq/Ollama) for text re-rendering, Stable Diffusion (local or via Replicate) for image synthesis, and Coqui TTS for audio — enabling fully functional deployment at near-zero marginal cost per student.

11. Layer 4 — The Feedback Loop

11.1 End-of-Lesson Preference Modal

At the completion of every content module, the learner is presented with a single, frictionless question: *"Which format worked best for you in this session?"* with three to four options corresponding to the formats used during that session (e.g., "The video explanations," "The simplified text," "The original content," "The practice tasks").

The learner's choice produces the Preference Delta — the primary ground-truth signal in the system. This label is fed back into the Orchestrator's replay buffer as the explicit reward signal for the format selection decisions made during that session. Over time, the Orchestrator learns to predict which format each individual learner will select before the learner has to express that preference explicitly.

The Preference Delta metric — the gap between the Orchestrator's predicted format and the learner's expressed preference — is the primary convergence benchmark for the POC evaluation. A Preference Delta trending toward zero across a cohort indicates that the system is successfully building accurate individual learner models.

11.2 The Energy Bar

The Energy Bar is a visible, always-accessible UI element within the learning interface. It takes the form of a small battery or gauge in the corner of the screen that the learner can drag down or tap to indicate that they need a break — regardless of what the Orchestrator has decided.

This element serves three functions:

- 1. Learner Agency.** The Energy Bar is the system's explicit acknowledgment that the learner is the authority on their own experience. No automated system, however well-designed, should override a student's self-knowledge of when they need to stop.
- 2. High-Magnitude Negative Reward.** When the Energy Bar is triggered, it fires a large negative reward signal into the Orchestrator's replay buffer, tagged to the

state vector at the moment of triggering. This teaches the Orchestrator that the sequence of signals leading up to that moment constitutes a pre-break trajectory — and that an earlier, lighter intervention (a nudge, a format switch) should have been applied before the learner reached the point of needing to manually halt.

3. Personalization of the Intervention Threshold. Students who trigger the Energy Bar frequently teach the Orchestrator to maintain a lower intervention threshold for their profile — intervening earlier in the signal progression — while students who rarely trigger it teach the Orchestrator that a more passive approach is appropriate for their profile.

11.3 Compounding Accuracy Over Time

The two feedback mechanisms together create a system that improves with every session. In the first session, the Orchestrator has no prior model of the individual learner and falls back to a population-level prior (the policy initialised on aggregate training data). By the third or fourth session, the combination of Preference Delta labels and Energy Bar events has produced enough per-learner signal to significantly personalise the policy. By the tenth session, the system's model of the individual learner should be substantially more accurate than any generic adaptation could be.

This compounding accuracy is the key long-term advantage of NeuroAdapt over static adaptive systems that assign students to fixed profiles (e.g., "visual learner," "auditory learner") and never update those assignments.

12. The Multi-Agent Network

NeuroAdapt's four layers can be understood as a network of specialised agents, each with a distinct responsibility and a defined interface to the others.

The Observer Agent is a passive, stateless data collection agent. It has no policy, no memory, and no decision-making responsibility. Its sole function is to compute and transmit the State Vector at each polling interval.

The Orchestrator Agent is the primary decision-making agent. It maintains a policy (the DDQN+VQC network), a replay buffer, and a per-learner session memory. It receives State Vectors and outputs Actions. It is the only agent that touches the reward function directly.

The Synthesis Agent is a reactive execution agent. It receives Action signals from the Orchestrator and produces the corresponding content variants. It has no policy of its own — it does not decide what to produce, only how to produce it given the Orchestrator's instruction.

The Feedback Agent mediates between the learner's explicit and implicit signals and the Orchestrator's policy update cycle. It collects Preference Delta labels and Energy Bar events and translates them into reward signals that are injected into the Orchestrator's replay buffer.

A potential future extension of this architecture includes a **Curriculum Discovery Agent** — a policy-gradient agent that operates at the session-planning level (rather than the within-session level), performing optimisation over the learner's historical digital footprint to sequence upcoming modules in the order most likely to produce stable engagement. This agent is not part of the initial NeuroAdapt implementation but represents a natural next layer of the multi-agent hierarchy.

13. End-to-End Session Flow

The following describes the complete sequence of events during a single NeuroAdapt learning session, from the moment the student opens a module to the moment the session ends.

Step 1 — Session Initialisation

The learner opens a content module via the LMS (Moodle, Canvas, or similar). The NeuroAdapt browser script initialises and begins passive observation. The Orchestrator is loaded with the learner's prior session model (or the population prior for a first session). The Synthesis Engine begins pre-fetching variants of the first content segment.

Step 2 — Baseline Observation (0–30 seconds)

The Observer records the first 30-second interval of signals. All five metrics are computed and assembled into the initial State Vector. No intervention is made in this window — this is the baseline observation period.

Step 3 — First Decision

At $T=30s$, the initial State Vector is transmitted to the Orchestrator. The DDQN+VQC policy computes Q-values for all six actions and selects the highest-value action. If Action 0 (Hold Course), no change occurs. If any other action is selected and the Q-value exceeds the 0.60 confidence threshold, the Synthesis Engine is triggered to deliver the appropriate format variant.

Step 4 — Continuous Adaptation

Steps 2 and 3 repeat every 30 seconds for the duration of the session. Each new State Vector is compared against the previous one to detect trajectories (e.g., a stall duration that has been steadily increasing over three consecutive intervals warrants a stronger intervention than a single isolated stall). The Orchestrator maintains a short-term session memory for this purpose.

Step 5 — Pre-fetch in Background

Simultaneously with Steps 2–4, the Synthesis Engine asynchronously generates variants of upcoming content segments based on the Orchestrator's current action distribution. The top-probability format for each upcoming segment is pre-rendered so that format transitions are seamless.

Step 6 — Energy Bar Event (if triggered)

If the learner triggers the Energy Bar at any point, the session is immediately paused, a break is offered, and the high-magnitude negative reward is logged. When the session resumes, the Orchestrator's policy for this learner is updated to apply earlier intervention in future sessions with similar signal trajectories.

Step 7 — Module Completion & Preference Modal

When the learner completes the module content, the end-of-lesson Preference Modal is displayed. The learner selects their preferred format. The Preference Delta is computed (predicted format vs. selected format) and logged. If the prediction was correct, a positive reward is applied to the corresponding state-action pairs. If incorrect, a negative reward is applied and the policy updates accordingly.

Step 8 — Session Model Update

The full session's state-action-reward trajectory is processed and the learner's individual model is updated. The updated model is stored (server-side, normalised, anonymised) and loaded at the start of the next session.

14. Expected Outcomes & Success Metrics

NeuroAdapt targets four primary measurable outcomes in the POC and institutional pilot phases.

Outcome 1 — Module Completion Rate

Target: 25% improvement in module completion rates for neurodivergent learners compared to the control group (static LMS).

Measurement: Percentage of learners who reach the end-of-module state and complete the Preference Modal, compared across the three arms of the evaluation (static LMS, classical DDQN, quantum-classical hybrid).

Evidence basis: Comparable agentic adaptive AI deployments in STEM education have demonstrated dropout rate reductions of up to 41%. A conservative 25% target is well within established benchmarks.

Outcome 2 — Assessment Performance

Target: 10–15% gain in post-module assessment scores for neurodivergent learners compared to the control group.

Measurement: Post-module quiz scores across the three evaluation arms.

Evidence basis: RL-based ITS systems have consistently produced double-digit percentage improvements in assessment performance. The 10–15% target reflects the expected contribution of format adaptation specifically, isolated from content difficulty adjustment.

Important note: NeuroAdapt achieves this not by making assessments easier. The

assessment content is held constant across all three arms. The performance gain comes entirely from improved knowledge encoding during the learning phase — because difficulty of decoding the content is no longer a competing cognitive load.

Outcome 3 — Preference Delta Convergence

Target: Preference Delta (gap between predicted and actual format preference) trending measurably toward zero across the pilot cohort, within 5–8 sessions per learner.

Measurement: Mean Preference Delta per learner, plotted across session number.

Evidence basis: The VQC's early-convergence advantage (Bhatia 2025; Cao 2025) supports rapid per-learner model formation. The three-arm ablation structure will directly measure whether the quantum layer produces faster convergence than the classical DDQN baseline.

Outcome 4 — Diagnosis-Agnostic Coverage

Target: Demonstrable improvement in outcomes for learners who self-report as neurodivergent but have no formal diagnosis, at levels comparable to those with formal diagnoses.

Measurement: Subgroup analysis within the treatment arms comparing diagnosed vs. undiagnosed neurodivergent learners on Outcomes 1 and 2.

Evidence basis: Because NeuroAdapt's signals are entirely behavioural and do not reference diagnostic categories, the system has no mechanism to produce different outcomes for diagnosed vs. undiagnosed learners with equivalent behavioural profiles. The evaluation will confirm this theoretically expected equivalence empirically.

15. Market Opportunity

Market Size

The global neurodivergent-focused EdTech market is valued at approximately USD 2.4 billion in 2025. It is projected to reach USD 6.4 billion by 2032, representing a compound annual growth rate of 15.1%. Personalised adaptive learning platforms are the fastest-growing segment within this market, ahead of assistive technology and specialised curriculum providers.

Asia Pacific, including India, is the highest-growth geographic region in this segment, driven by rapidly expanding internet access, government EdTech investment, and growing societal awareness of neurodivergent conditions.

Competitive Landscape

Current competitors fall into three broad categories:

Assistive technology tools (Read&Write, ClaroRead, NaturalReader): These provide text-to-speech, screen reading, and dyslexia-friendly formatting. They are static, opt-in features that require the user to configure them manually. They do not adapt in real time.

General adaptive EdTech platforms (Knewton, Khan Academy, Coursera with personalisation): These adapt difficulty and content sequencing based on performance metrics. They are designed for the general population and optimise for correctness, not cognitive stability. They do not target neurodivergent-specific signals.

Clinical digital therapeutics (EndeavorRx, Mightier): These are FDA-cleared medical devices that are prescribed by clinicians, require parental/clinical involvement, and are used in therapeutic contexts rather than mainstream academic settings. They are not integrated into LMS environments.

NeuroAdapt's position is unique: It is the only system that combines real-time behavioural cognitive state sensing, RL-based policy optimisation with a stability-focused reward, and multimodal generative content synthesis, in a diagnosis-agnostic, LMS-integrated, B2B SaaS package. No existing product occupies

this position.

Economic Burden Context

ADHD alone is associated with a USD 122.8 billion annual societal excess cost in the United States, of which USD 96 billion represents workforce productivity loss and unemployment traceable to lower educational attainment. Dyslexia accounts for an estimated additional USD 200 billion in annual economic impact across the OECD. The argument for investment in upstream educational intervention is not merely moral — it is economically overwhelming.

16. Cost Architecture & ROI

Infrastructure Costs

The Observer: Runs entirely in the browser. Zero server cost. Zero marginal cost per additional learner. All five signals are computed on-device using standard browser JavaScript with no dependencies on external APIs.

The Orchestrator: Inference on the DDQN+VQC network is computationally minimal. The state vector has five dimensions; the action space has six options. Forward pass time on a standard CPU server is measured in milliseconds. The VQC, during the POC phase, runs on free quantum simulators (PennyLane default simulator). During institutional pilot, it migrates to cloud-based NISQ simulators at low cost. Real quantum hardware is a Phase 3 consideration.

The Synthesis Engine: This is the only component with significant variable cost. GPT-4o API pricing at current rates produces a moderate per-session cost when re-rendering is triggered. HeyGen avatar video is more expensive per generation. ElevenLabs TTS has minimal cost per character.

The 60% confidence gate dramatically reduces actual API call frequency. In sessions where the learner is well-matched to their content, the vast majority of 30-second intervals result in Action 0 (Hold Course) with high confidence, triggering no API calls. Synthesis costs are essentially proportional to session difficulty —

the sessions where the cost is incurred are precisely the sessions where the intervention produces the highest value.

Open-source substitution: Every paid API in the Synthesis Engine has a working open-source substitute for zero-cost deployments:

- GPT-4o → Llama-3 (via Ollama local hosting or Groq free tier)
- HeyGen → Stable Diffusion + D-ID or SadTalker (open-source)
- ElevenLabs → Coqui TTS (open-source, self-hosted)

ROI Case

The ROI argument is straightforward. In higher education, a failed course means:

- The student pays for re-enrolment (in India, typically Rs. 30,000–80,000 per credit in private universities, and indirect costs in public ones).
- The institution loses completion-rate funding tied to student graduation metrics.
- The student loses one to two semesters of workforce entry time.
- Mental health costs from repeated academic failure compound.

NeuroAdapt's full annual per-student operating cost — including API calls, server infrastructure, and model training — at reasonable usage levels is a small fraction of the cost of a single failed course. Preventing even a marginal improvement in completion rates produces a return on investment that is straightforward to demonstrate